MRV for the scalar Signorini problem:
constrained postprocessing, a posteriori estimator and experiments

Working note

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Abstract

We extend the MRV scheme [10], which combines Stenberg's postprocessing [12] with local residual minimization, to the scalar Signorini problem. There are three new ingredients. The first is a minres postprocessing with a positivity constraint on Γ_C, imposed on Bernstein coefficients; the trial still coincides with Stenberg's, now constrained, and the indicator $\tilde{\eta}_K$ absorbs the cost of the constraint. The second is a conforming, admissible reconstruction v_h that sets the trace to zero on discrete contact edges wherever this is cheaper than paying for complementarity. The third is a boundary term $\eta_{C,E}$ on Γ_C. Reliability follows from a Prager–Synge type identity for Signorini, without saturation and without a Fortin operator, and is stated in the MRV norm with jumps. We add the reaction case, discrete complementarity via KKT, and experiments with RT_0, BDM_1 and BDM_2, including a numerical verification of the identity. Everything labelled as an *order count* is heuristic and not proved.

Changes with respect to version 1.

- **Correction:** the oscillation of f is *not* of higher order for BDM_p; it is of the same order as the error (Remark 6.2). It is used for marking together with η.
- The error is measured in the MRV norm with jumps. Efficiency of the jump and Dirichlet terms now holds with constant 1 (Section 8).
- Coefficient-wise discrete complementarity via KKT (Proposition 2.1) and choice of solver.
- A local divergence-free correction when the sign of $\boldsymbol{q}_h \cdot \boldsymbol{n}$ is only imposed in moments (Remark 2.4).
- Full proof of Lemma 4.3(d).
- Reaction case (Section 9) and numerical experiments (Section 10).

Changes in version 3.

- Prototype extended to BDM_p and experiments with BDM_2 ($k = 3$), including BGK's example 7.2 (Section 10.1).
- Numerical verification of the identity with the full reconstruction v_h and an empirical estimate of $C_{\mathcal{A}}$ (Table 17).
- Remark 7.2: why the norm carries no trace term on Γ_C.
- Refined product bound on transition edges (Proposition 8.1(iv)) and comparison with Krause–Veeser–Walloth (Remark 8.3).
- Stenberg's full constraint in the postprocessing for reaction with $m \geq 1$ (Section 9), used for FHS 4.2 with BDM_2.

1 Problem and notation

Let $\Omega \subset \mathbb{R}^2$ be a polygon with $\partial\Omega = \overline{\Gamma}_D \cup \overline{\Gamma}_C$, $\Gamma_D \cap \Gamma_C = \emptyset$ and $|\Gamma_D| > 0$, and let $f \in L^2(\Omega)$. The problem is

$$-\Delta u = f \text{ in } \Omega, \qquad u = 0 \text{ on } \Gamma_D, \qquad u \geq 0, \quad \partial_n u \geq 0, \quad u\,\partial_n u = 0 \text{ on } \Gamma_C, \tag{1}$$

with $\boldsymbol{n}$ the outward normal. Let $H^1_D := \{v \in H^1(\Omega) : v|_{\Gamma_D} = 0\}$ and $\mathcal{K} := \{v \in H^1_D : v|_{\Gamma_C} \geq 0\}$. The weak formulation reads: find $u \in \mathcal{K}$ such that

$$(\nabla u, \nabla(v-u)) \geq (f, v-u) \qquad \forall v \in \mathcal{K}. \tag{2}$$

Let $\boldsymbol{q} := -\nabla u$, so that $\operatorname{div} \boldsymbol{q} = f$. For $\boldsymbol{\tau} \in H(\operatorname{div}, \Omega)$ and $w \in H^1_D$ we define

$$\langle \boldsymbol{\tau} \cdot \boldsymbol{n}, w \rangle_{\Gamma_C} := (\boldsymbol{\tau}, \nabla w) + (\operatorname{div} \boldsymbol{\tau}, w). \tag{3}$$

This expression depends only on $w|_{\Gamma_C}$, and if $\boldsymbol{\tau} \cdot \boldsymbol{n} \in L^2(\Gamma_C)$ it coincides with $(\boldsymbol{\tau} \cdot \boldsymbol{n}, w)_{\Gamma_C}$.

Lemma 1.1. *$\langle \boldsymbol{q} \cdot \boldsymbol{n}, u \rangle_{\Gamma_C} = 0$ and $\langle -\boldsymbol{q} \cdot \boldsymbol{n}, v \rangle_{\Gamma_C} \geq 0$ for all $v \in \mathcal{K}$.*

Proof. By (3), $\langle \boldsymbol{q} \cdot \boldsymbol{n}, w \rangle_{\Gamma_C} = -(\nabla u, \nabla w) + (f, w)$, so (2) is equivalent to $\langle -\boldsymbol{q} \cdot \boldsymbol{n}, v - u \rangle_{\Gamma_C} \geq 0$ for all $v \in \mathcal{K}$. Taking $v = 0$ and $v = 2u$ gives $\langle \boldsymbol{q} \cdot \boldsymbol{n}, u \rangle_{\Gamma_C} = 0$, and the second claim follows. □

Let $\mathcal{T}_h$ be a shape-regular triangulation compatible with the partition Γ_D/Γ_C. We write $\mathcal{F}^i_h, \mathcal{F}^D_h, \mathcal{F}^C_h$ for the interior edges, the edges on Γ_D and the edges on Γ_C, and $\mathcal{F}^i_K, \mathcal{F}^D_K, \mathcal{F}^C_K$ for those of an element K. For $E \in \mathcal{F}^C_h$, K_E is the unique element containing it. We use $(\cdot,\cdot)_E$ for the $L^2(E)$ inner product.

Remark 1.2 (Neumann and obstacle)**.** With a homogeneous Neumann part Γ_N ($\boldsymbol{q} \cdot \boldsymbol{n} = 0$) and an obstacle $u \geq \chi$ on Γ_C, everything below holds after fixing $\boldsymbol{q}_h \cdot \boldsymbol{n} = 0$ on Γ_N and replacing u, ν_h and v_h by $u - \chi$, $\nu_h - \mathcal{I}\chi$ and $v_h - \mathcal{I}\chi$ on Γ_C, with χ piecewise linear on the mesh of Γ_C. There is no boundary term on Γ_N, since $(\boldsymbol{q} - \boldsymbol{q}_h) \cdot \boldsymbol{n} = 0$ there. This is how BGK's example 7.3 is implemented in Section 10.

2 Mixed solution

Let $H_h = \mathrm{BDM}_p$ with $p \geq 1$, as in [10], or $H_h = \mathrm{RT}_0$. We write $V_h := \operatorname{div} H_h = \mathbb{P}^m(\mathcal{T}_h)$, with $m = p - 1$ or $m = 0$, and Q_h for the L^2 projection onto V_h. The degree of the postprocessing is $k = p + 1$ for BDM_p and $k = 1$ for RT_0. For $E \in \mathcal{F}^C_h$, let $b_{E,i}(g)$, $i = 0, \ldots, p$, be the degree-p Bernstein coefficients of $g \in \mathbb{P}^p(E)$ (for RT_0, $p := 0$).

(H1) $\boldsymbol{q}_h \in H_h$, $\operatorname{div} \boldsymbol{q}_h = Q_h f$ and $\boldsymbol{q}_h \cdot \boldsymbol{n}|_E \leq 0$ pointwise for every $E \in \mathcal{F}^C_h$.

(H3) *(Used only in the remarks on efficiency.)* Exact discrete complementarity, in the sense of Proposition 2.1.

Hypothesis (H2) is stated in Section 3, because it holds by construction.

One way to obtain (H1) is the discrete mixed variational inequality: find $(\boldsymbol{q}_h, u_h) \in H^B_h \times V_h$, with $H^B_h := \{\boldsymbol{\tau} \in H_h : b_{E,i}(\boldsymbol{\tau} \cdot \boldsymbol{n}|_E) \leq 0 \ \forall E \in \mathcal{F}^C_h, \ \forall i\}$, such that

$$(\boldsymbol{q}_h, \boldsymbol{\tau} - \boldsymbol{q}_h) - (u_h, \operatorname{div}(\boldsymbol{\tau} - \boldsymbol{q}_h)) \geq 0 \quad \forall \boldsymbol{\tau} \in H^B_h, \qquad (\operatorname{div} \boldsymbol{q}_h, w) = (f, w) \quad \forall w \in V_h. \tag{4}$$

Since the Bernstein basis is nonnegative, $\boldsymbol{\tau} \in H^B_h$ implies $\boldsymbol{\tau} \cdot \boldsymbol{n} \leq 0$ pointwise on Γ_C. For RT_0 and BDM_1 the condition is also necessary: the normal trace is constant or linear on E, and its Bernstein coefficients are its endpoint values. Reliability uses only (H1): the estimator does not depend on how $\boldsymbol{q}_h$ was obtained. Of u_h, only its element means are used.

Proposition 2.1 (discrete complementarity)**.** *Let $(\boldsymbol{q}_h, u_h)$ solve (4). There exist $\mu_{E,i} \geq 0$ such that*

$$(\boldsymbol{q}_h, \boldsymbol{\tau}) - (u_h, \operatorname{div} \boldsymbol{\tau}) + \sum_{E \in \mathcal{F}^C_h} \sum_i \mu_{E,i}\, b_{E,i}(\boldsymbol{\tau} \cdot \boldsymbol{n}|_E) = 0 \quad \forall \boldsymbol{\tau} \in H_h, \qquad \mu_{E,i}\, b_{E,i}(\boldsymbol{q}_h \cdot \boldsymbol{n}|_E) = 0 \quad \forall E, i.$$

Proof. Let $\ell(\boldsymbol{\tau}) := (\boldsymbol{q}_h, \boldsymbol{\tau}) - (u_h, \operatorname{div}\boldsymbol{\tau})$. The inequality in (4) says $\ell(\boldsymbol{\tau}) \geq \ell(\boldsymbol{q}_h)$ for all $\boldsymbol{\tau} \in H_h^B$. Since H_h^B is a polyhedral convex cone, taking $\boldsymbol{\tau} = 0$ and $\boldsymbol{\tau} = 2\boldsymbol{q}_h$ gives $\ell(\boldsymbol{q}_h) = 0$, and then $\ell \geq 0$ on H_h^B. By Farkas' lemma, ℓ lies in the cone generated by the linear forms $-b_{E,i}(\cdot\,\cdot\boldsymbol{n})$, that is, $\ell = -\sum \mu_{E,i} b_{E,i}(\cdot\,\cdot\boldsymbol{n})$ with $\mu_{E,i} \geq 0$. Finally, $0 = \ell(\boldsymbol{q}_h) = -\sum \mu_{E,i} b_{E,i}(\boldsymbol{q}_h \cdot \boldsymbol{n})$, and each summand $\mu_{E,i} b_{E,i}(\boldsymbol{q}_h \cdot \boldsymbol{n})$ is ≤ 0, so all of them vanish. □

Remark 2.2 (interpretation and what is missing)**.** Comparing with the continuous identity $(\boldsymbol{q}, \boldsymbol{\tau}) - (u, \operatorname{div}\boldsymbol{\tau}) + \langle \boldsymbol{\tau}\cdot\boldsymbol{n}, u\rangle_{\Gamma_C} = 0$, the term $\sum_i \mu_{E,i} b_{E,i}(\boldsymbol{\tau}\cdot\boldsymbol{n})$ plays the role of $\int_E u\,\boldsymbol{\tau}\cdot\boldsymbol{n} = \sum_i b_{E,i}(\boldsymbol{\tau}\cdot\boldsymbol{n}) \int_E u B_{E,i}$. Thus $\mu_{E,i}$ approximates $\int_E u B_{E,i}$, which is positive as soon as $u > 0$ somewhere on E. If the multipliers converge to this limit, then on edges of actual separation all $\mu_{E,i}$ are positive and $\boldsymbol{q}_h \cdot \boldsymbol{n} \equiv 0$; that is, $\mathcal{C}_h$ contains only edges on which u vanishes on a part. This convergence requires an a priori analysis of the multipliers that I do not have. In the experiments, the number of misclassified edges is at most 3 per level, always within distance $2h$ of a transition.

Remark 2.3 (solver)**.** We use the primal-dual active set strategy, which is a semismooth Newton method [5]. The active set is $\mathcal{A} = \{(E,i) : \mu_{E,i} + \kappa_E\, b_{E,i}(\boldsymbol{q}_h\cdot\boldsymbol{n}) > 0\}$ with $\kappa_E = h_E$. On $\mathcal{A}$ one imposes $b_{E,i}(\boldsymbol{q}_h\cdot\boldsymbol{n}) = 0$ and off $\mathcal{A}$ one imposes $\mu_{E,i} = 0$; each iteration is a linear saddle-point system. In the experiments it terminated in 1 to 20 iterations, with KKT residual at machine precision, so the complementarity of Proposition 2.1 holds exactly for the computed solution.

Remark 2.4 (sign imposed only in moments)**.** If the mixed solver only guarantees $\int_E \boldsymbol{q}_h\cdot\boldsymbol{n} \leq 0$ on each $E \in \mathcal{F}_h^C$, (H1) may fail. In that case no equilibration across elements is needed, only a local divergence-free correction.

- Let a be the vertex of K_E opposite E and b, c the endpoints of E.
- Let $g_E^* \in \mathbb{P}^p(E)$ be the polynomial with Bernstein coefficients ≤ 0, with the same mean as $\boldsymbol{q}_h\cdot\boldsymbol{n}|_E$, closest to it in $L^2(E)$. This is a small QP. Let $g_E := g_E^* - \boldsymbol{q}_h\cdot\boldsymbol{n}|_E$, which has zero mean.
- The antiderivative of g_E along E vanishing at b also vanishes at c, so it can be written as $(\lambda_b\lambda_c)|_E\,\psi_E$ with $\psi_E \in \mathbb{P}^{p-1}(E)$. Extend ψ_E to K_E as a homogeneous polynomial in (λ_b, λ_c).
- Define $\boldsymbol{\delta}_E := \operatorname{curl}(\lambda_b\lambda_c\psi_E)$ on K_E, oriented so that $\boldsymbol{\delta}_E\cdot\boldsymbol{n} = g_E$ on E, and $\boldsymbol{\delta}_E := 0$ outside K_E.

Then $\boldsymbol{\delta}_E \in \mathbb{P}^p(K_E)^2 \subset \mathrm{BDM}_p(K_E)$, its normal trace vanishes on the other two edges, and $\operatorname{div}\boldsymbol{\delta}_E = 0$. Hence $\boldsymbol{\tau} := \boldsymbol{q}_h + \sum_E \boldsymbol{\delta}_E$ satisfies (H1) with $\operatorname{div}\boldsymbol{\tau} = Q_h f$. Theorem A holds with $\boldsymbol{\tau}$ in place of $\boldsymbol{q}_h$, adding the computable term $(\sum_E \|\boldsymbol{\delta}_E\|_{K_E}^2)^{1/2}$, of the order of $h_E^{1/2}\|g_E\|_E$, to the estimator. For RT_0 it is not needed, because $\boldsymbol{q}_h\cdot\boldsymbol{n}|_E$ is constant.

3 Constrained MRV postprocessing

Bernstein. Let a_1, a_2, a_3 be the vertices of K and λ_i its barycentric coordinates. For $|\alpha| = k$ we define $B_\alpha := \frac{k!}{\alpha!}\lambda^\alpha$, and we write $w = \sum_\alpha c_\alpha(w) B_\alpha$ for $w \in \mathbb{P}^k(K)$. For $E \in \mathcal{F}_K^C$ with opposite vertex a_E, let $I_E := \{\alpha : \alpha_{a_E} = 0\}$, which has $k+1$ elements. Since $\lambda_{a_E} = 0$ on E, we have $w|_E = \sum_{\alpha\in I_E} c_\alpha(w) B_\alpha|_E$, and $\{B_\alpha|_E\}$ is the 1D Bernstein basis of degree k on E. Therefore,

$$c_\alpha(w) \geq 0 \;\; \forall \alpha \in I_E \quad\Longrightarrow\quad w|_E \geq 0.$$

The converse fails for $k \geq 2$. For instance, $(2s-1)^2$ on $[0,1]$ has coefficients $(1,-1,1)$. Moreover, $\int_K B_\alpha = |K|/N$ with $N = \frac{(k+1)(k+2)}{2}$, so $(w,1)_K = \frac{|K|}{N}\sum_\alpha c_\alpha(w)$.

Local problem. Let $U_K := \mathbb{P}^k(K)$, $V_K := \mathbb{P}_*^{k+1}(K) := \{v \in \mathbb{P}^{k+1}(K) : (v,1)_K = 0\}$,

$$U_K^+ := \{w \in U_K :\; (w,1)_K = (u_h,1)_K,\;\; c_\alpha(w) \geq 0 \;\; \forall \alpha \in I_E,\; \forall E \in \mathcal{F}_K^C\},$$

and $\|\boldsymbol{\tau}\|_{*,K} := \sup_{v\in V_K\setminus\{0\}} (\boldsymbol{\tau}, \nabla v)_K / \|\nabla v\|_K$. If $\mathcal{F}_K^C = \emptyset$, U_K^+ carries only the mean constraint, as in [10, (18)]. We define

$$\nu_K := \operatorname*{arg\,min}_{w\in U_K^+} \tfrac{1}{2}\|\boldsymbol{q}_h + \nabla w\|_{*,K}^2, \qquad (\nabla\varepsilon_K, \nabla v)_K = -(\boldsymbol{q}_h + \nabla\nu_K, \nabla v)_K \quad \forall v \in V_K, \tag{5}$$

with $\varepsilon_K \in V_K$ and $\tilde{\eta}_K := \|\nabla \varepsilon_K\|_K$. We write $\nu_h|_K := \nu_K$. In saddle-point form: find $(\varepsilon_K, \nu_K) \in V_K \times U_K^+$ such that

$$(\nabla \varepsilon_K, \nabla v)_K + (\nabla \nu_K, \nabla v)_K = -(\boldsymbol{q}_h, \nabla v)_K \ \ \forall v \in V_K, \qquad (\nabla \varepsilon_K, \nabla(w - \nu_K))_K \le 0 \ \ \forall w \in U_K^+. \tag{6}$$

If $\mathcal{F}_K^C = \emptyset$, $w - \nu_K$ ranges over $\mathbb{P}_*^k(K)$, the inequality becomes an equality and [10, (19)] is recovered.

(H2) *(By construction.)* $c_\alpha(\nu_K) \ge 0$ for all $\alpha \in I_E$ and all $E \in \mathcal{F}_K^C$, and $(\nu_K, 1)_K = (u_h, 1)_K$.

Lemma 3.1. *Let Π_K be the $L^2(K)$ projection onto ∇V_K. Assume that K has at most one edge on Γ_C, or that $k \ge 2$.*

(a) $U_K^+ \ne \emptyset$ and (5) has a unique solution.

(b) Coincidence: $\nu_K = \arg\min_{w \in U_K^+} \|\boldsymbol{q}_h + \nabla w\|_K$. *This is Stenberg's postprocessing with a constraint.*

(c) $\nabla \varepsilon_K = -\Pi_K(\boldsymbol{q}_h + \nabla \nu_K)$. In particular, $\tilde{\eta}_K = \|\boldsymbol{q}_h + \nabla \nu_K\|_{*,K} \le \|\boldsymbol{q}_h + \nabla \nu_K\|_K$.

(d) (ε_K, ν_K) satisfies (6).

(e) Let $\nu_{K,0}$ be the unconstrained MRV postprocessing, which coincides with Stenberg's [10, Prop. 3.1], and let $\tilde{\eta}_{K,0}$ be its indicator. Then

$$\tilde{\eta}_K^2 = \tilde{\eta}_{K,0}^2 + \|\nabla(\nu_K - \nu_{K,0})\|_K^2.$$

Proof. For $w \in U_K$ we have $w - \bar{w}_K \in \mathbb{P}_*^k(K) \subset V_K$, so $\nabla U_K \subset \nabla V_K$. Hence $\|\boldsymbol{q}_h + \nabla w\|_{*,K} = \|\Pi_K \boldsymbol{q}_h + \nabla w\|_K$.

(a) If $\mathcal{F}_K^C = \{E\}$, let $w = 3\bar{u}_h \lambda_{a_E}$. Since $\lambda_{a_E} = \lambda_{a_E}(\lambda_1 + \lambda_2 + \lambda_3)^{k-1}$, only multi-indices with $\alpha_{a_E} \ge 1$ appear, and $c_\alpha(w) = 0$ on I_E. Moreover $(3\lambda_{a_E}, 1)_K = |K|$. If there are two edges E_1, E_2 and $k \ge 2$, we use $w = 12\bar{u}_h \lambda_{a_{E_1}} \lambda_{a_{E_2}}$, with $(\lambda_i \lambda_j, 1)_K = |K|/12$. The functional $w \mapsto \frac{1}{2}\|\Pi_K \boldsymbol{q}_h + \nabla w\|_K^2$ is strictly convex on the set of fixed mean, since ∇ is injective there, and it is coercive. The set U_K^+ is convex, closed and nonempty, in finite dimension.

(b) By Pythagoras, $\|\boldsymbol{q}_h + \nabla w\|_K^2 = \|(I - \Pi_K)\boldsymbol{q}_h\|_K^2 + \|\Pi_K \boldsymbol{q}_h + \nabla w\|_K^2$, and the first term does not depend on w.

(c) By (5), $\nabla \varepsilon_K$ is the projection of $-(\boldsymbol{q}_h + \nabla \nu_K)$ onto ∇V_K, and Π_K is a contraction.

(d) Let $J(w) := \frac{1}{2}\|\Pi_K \boldsymbol{q}_h + \nabla w\|_K^2$ and $\boldsymbol{t} := -\Pi_K \boldsymbol{q}_h$. The optimality condition on the convex set is $(\nabla \nu_K - \boldsymbol{t}, \nabla(w - \nu_K))_K \ge 0$, and by (c) we have $\nabla \varepsilon_K = \boldsymbol{t} - \nabla \nu_K$.

(e) Let $S := \nabla U_K$ (a subspace of ∇V_K), $C := \nabla U_K^+ \subset S$ and P_S the projection onto S. For $\boldsymbol{c} \in S$, $\|\boldsymbol{t} - \boldsymbol{c}\|^2 = \|\boldsymbol{t} - P_S \boldsymbol{t}\|^2 + \|P_S \boldsymbol{t} - \boldsymbol{c}\|^2$. We have $\nabla \nu_{K,0} = P_S \boldsymbol{t}$ and $\tilde{\eta}_{K,0} = \|\boldsymbol{t} - P_S \boldsymbol{t}\|$. Moreover, $\nabla \nu_K$ minimizes $\|\boldsymbol{t} - \boldsymbol{c}\|$ over C, and $\tilde{\eta}_K = \|\boldsymbol{t} - \nabla \nu_K\|$. □

Remark 3.2 (computation)**.** Only elements with an edge on Γ_C require anything new. By (b), it suffices to solve the strictly convex QP

$$\min_{c \in \mathbb{R}^N} \ \tfrac{1}{2} c^\top A c - g^\top c, \qquad A_{\alpha\beta} = (\nabla B_\alpha, \nabla B_\beta)_K, \quad g_\alpha = -(\boldsymbol{q}_h, \nabla B_\alpha)_K,$$

subject to $\sum_\alpha c_\alpha = N\bar{u}_h$ and $c_\alpha \ge 0$ for $\alpha \in \bigcup_E I_E$. It has $N = \frac{(k+1)(k+2)}{2}$ unknowns and $k+1$ bounds per edge. If the unconstrained Stenberg postprocessing is already admissible, $\nu_K = \nu_{K,0}$. Otherwise one eliminates the mean with a coefficient outside $\bigcup_E I_E$ and solves a bound-constrained linear least-squares problem (in the prototype, `scipy.optimize.lsq_linear` with the `bvls` method). Then ε_K follows from the linear system (5).

Remark 3.3 (feasibility)**.** For $k = 1$ (RT_0) and an element with two edges on Γ_C, U_K^+ may be empty. For instance, $u = -xy$ at a corner with contact on both axes has negative mean. In that case we require, as a mesh condition, that no element have two edges on Γ_C. For BDM_p we have $k \ge 2$ and this is not needed. The meshes of Section 10 satisfy the condition, and newest-vertex bisection preserves it.

4 Discrete contact set and reconstruction

Definition 4.1. $\mathcal{C}_h := \{E \in \mathcal{F}_h^C : \ \boldsymbol{q}_h \cdot \boldsymbol{n}|_E \not\equiv 0\}$.

Let $\mathcal{D}_h$ be the set of domain points of degree k, that is, the points $\sum_i \frac{\alpha_i}{k} a_i$. A function in $\mathbb{P}^k(\mathcal{T}_h) \cap C^0(\overline{\Omega})$ is determined by single-valued Bernstein coefficients c_x, $x \in \mathcal{D}_h$. For $x \in \mathcal{D}_h$ we define $\mathcal{T}_x := \{K : x \in K\}$ and $\mathcal{T}_x^C := \{K \in \mathcal{T}_x : x \in \overline{E} \text{ for some } E \in \mathcal{F}_K^C\}$.

Definition 4.2. For $Z \subset \mathcal{F}_h^C$, let $\mathcal{A}_Z \nu_h \in \mathbb{P}^k(\mathcal{T}_h) \cap H^1(\Omega)$ be the function with coefficients

$$c_x := \begin{cases} 0, & x \in \overline{\Gamma}_D \text{ or } x \in \overline{E} \text{ for some } E \in Z, \\ \frac{1}{\#\mathcal{T}_x^C} \sum_{K \in \mathcal{T}_x^C} c_x(\nu_K), & x \in \overline{\Gamma}_C \text{ (otherwise)}, \\ \frac{1}{\#\mathcal{T}_x} \sum_{K \in \mathcal{T}_x} c_x(\nu_K), & \text{otherwise.} \end{cases}$$

Let $v_h^0 := \mathcal{A}_\emptyset \nu_h$. For $E \in \mathcal{C}_h$ we define

$$\eta_{C,E}^2 := \min\{h_E^{-1}\|\nu_h\|_E^2,\ 2(v_h^0, -\boldsymbol{q}_h \cdot \boldsymbol{n})_E\}, \qquad \mathcal{Z}_h := \{E \in \mathcal{C}_h : \ h_E^{-1}\|\nu_h\|_E^2 \le 2(v_h^0, -\boldsymbol{q}_h \cdot \boldsymbol{n})_E\}, \tag{7}$$

and $v_h := \mathcal{A}_{\mathcal{Z}_h} \nu_h$.

Lemma 4.3. *Under (H1) and (H2):*

(a) $v_h^0, v_h \in \mathbb{P}^k(\mathcal{T}_h) \cap H_D^1$.

(b) $v_h^0 \ge 0$ *and* $v_h \ge 0$ *on* Γ_C*;* $v_h = 0$ *on each* $E \in \mathcal{Z}_h$*; and* $0 \le v_h \le v_h^0$ *on* Γ_C*. In particular,* $v_h \in \mathcal{K}$.

(c) $(v_h, -\boldsymbol{q}_h \cdot \boldsymbol{n})_{\Gamma_C} \le \frac{1}{2} \sum_{E \in \mathcal{C}_h \setminus \mathcal{Z}_h} \eta_{C,E}^2$.

(d) $\|\nabla_h(\nu_h - v_h)\| \le C_{\mathcal{A}}\, J$ *and* $\|\nu_h - v_h\| \le C_{\mathcal{A}}\, h\, J$*, where* $C_{\mathcal{A}}$ *depends only on shape regularity and on* k*,* $h := \max_K h_K$*, and*

$$J^2 := \sum_{F \in \mathcal{F}_h^i} h_F^{-1}\|[\nu_h]\|_F^2 + \sum_{F \in \mathcal{F}_h^D} h_F^{-1}\|\nu_h\|_F^2 + \sum_{E \in \mathcal{Z}_h} h_E^{-1}\|\nu_h\|_E^2.$$

Proof. (a) Immediate by construction.

(b) Let $E \in \mathcal{F}_h^C$. The coefficients of $v_h|_E$ at the interior points of E are $c_x(\nu_{K_E}) \ge 0$, by (H2). At the endpoints of E they are either 0 or an average of values $c_x(\nu_{K'})$ with $K' \in \mathcal{T}_x^C$, all ≥ 0 by (H2), since x lies on an edge of K' contained in Γ_C. Since the $B_\alpha|_E$ are ≥ 0, it follows that $v_h|_E \ge 0$, and likewise for v_h^0. Each coefficient of v_h equals that of v_h^0 or is zero, so $v_h \le v_h^0$.

(c) On $E \in \mathcal{Z}_h$ we have $v_h = 0$. On $E \in \mathcal{F}_h^C \setminus \mathcal{C}_h$ we have $\boldsymbol{q}_h \cdot \boldsymbol{n} \equiv 0$. On $E \in \mathcal{C}_h \setminus \mathcal{Z}_h$, by (b) and since $-\boldsymbol{q}_h \cdot \boldsymbol{n} \ge 0$, we get $(v_h, -\boldsymbol{q}_h \cdot \boldsymbol{n})_E \le (v_h^0, -\boldsymbol{q}_h \cdot \boldsymbol{n})_E = \frac{1}{2}\eta_{C,E}^2$.

(d) Let $K \in \mathcal{T}_h$ and $d_x := c_x(\nu_K) - c_x(v_h)$ for $x \in \mathcal{D}_K$. By norm equivalence on $\mathbb{P}^k(\hat{K})$ and scaling (in 2D the H^1 seminorm is invariant), with C depending only on shape regularity and on k,

$$\|\nabla(\nu_K - v_h)\|_K^2 \le C \sum_{x \in \mathcal{D}_K} d_x^2, \qquad \|\nu_K - v_h\|_K^2 \le C\, h_K^2 \sum_{x \in \mathcal{D}_K} d_x^2.$$

If x is interior to K, then $\mathcal{T}_x = \{K\}$ and $d_x = 0$. If $x \in \partial K$ is not a zeroed point, $c_x(v_h)$ is an average of values $c_x(\nu_{K'})$ with $K' \in \mathcal{T}_x$ (or $\mathcal{T}_x^C \subset \mathcal{T}_x$), and $|d_x| \le \max_{K' \in \mathcal{T}_x} |c_x(\nu_K) - c_x(\nu_{K'})|$. The patch $\mathcal{T}_x$ is connected through faces containing x: there is a chain $K = K_0, \dots, K_r = K'$ with $F_j := K_{j-1} \cap K_j \ni x$ and r bounded by shape regularity. Since $c_x(\nu_{K_{j-1}}) - c_x(\nu_{K_j})$ is the Bernstein coefficient at x of $[\nu_h]|_{F_j} \in \mathbb{P}^k(F_j)$, norm equivalence on $\mathbb{P}^k(F_j)$ gives

$$|c_x(\nu_{K_{j-1}}) - c_x(\nu_{K_j})| \le C\, h_{F_j}^{-1/2} \|[\nu_h]\|_{F_j}.$$

If x is a zeroed point, there is $K' \in \mathcal{T}_x$ with an edge $F' \in \mathcal{F}_h^D \cup \mathcal{Z}_h$ such that $x \in \overline{F'}$, and $|d_x| \le |c_x(\nu_K) - c_x(\nu_{K'})| + |c_x(\nu_{K'})|$ with $|c_x(\nu_{K'})| \le C\, h_{F'}^{-1/2} \|\nu_h\|_{F'}$. Summing over $x \in \mathcal{D}_K$ and $K \in \mathcal{T}_h$, each edge appears a bounded number of times, which gives both bounds. Restricting the average to $\mathcal{T}_x^C$ only changes which elements of $\mathcal{T}_x$ are averaged. □

5 Indicator

Definition 5.1. For $K \in \mathcal{T}_h$, let

$$\eta_K^2 := \tilde{\eta}_K^2 + \|\boldsymbol{q}_h + \nabla \nu_K\|_K^2 + \tfrac{1}{2} \sum_{F \in \mathcal{F}_K^i} h_F^{-1} \|[\nu_h]\|_F^2 + \sum_{F \in \mathcal{F}_K^D} h_F^{-1} \|\nu_h\|_F^2 + \sum_{E \in \mathcal{F}_K^C \cap \mathcal{C}_h} \eta_{C,E}^2,$$

and $\eta^2 := \sum_K \eta_K^2$. Separately, as a data term, we define

$$\mathrm{osc}^2 := \sum_K \mathrm{osc}_K^2, \qquad \mathrm{osc}_K := \frac{h_K}{\pi} \|f - Q_h f\|_K. \tag{8}$$

Marking uses $\eta_K^2 + \mathrm{osc}_K^2$ (Remark 6.2).

By (7), $\eta_{C,E}^2 = h_E^{-1} \|\nu_h\|_E^2$ on $\mathcal{Z}_h$. Hence, with $S^2 := \sum_{E \in \mathcal{C}_h \setminus \mathcal{Z}_h} \eta_{C,E}^2$,

$$\eta^2 \;\geq\; \|\boldsymbol{q}_h + \nabla_h \nu_h\|^2 + J^2 + S^2. \tag{9}$$

Term	MRV, estimator (25)	Signorini (this note)
$\tilde{\eta}_K = \|\nabla \varepsilon_K\|_K$	same	constrained; Lemma 3.1(e)
$\|\boldsymbol{q}_h + \nabla \nu_K\|_K$	same	same
jumps $h_F^{-1} \|[\nu_h]\|_F^2$	same	same
Dirichlet boundary $h_F^{-1} \|u_D - \nu_h\|_F^2$	same	same (here $u_D = 0$)
contact boundary	absent	$\eta_{C,E}$, (7)
oscillation	$\mathrm{osc}(\boldsymbol{q})$, (23), via Fortin	osc, (8), from $f - \operatorname{div} \boldsymbol{q}_h$; used for marking

Table 1: Comparison with the improved estimator (25) of [10].

Computing the estimator.

1. Solve the mixed problem to obtain $\boldsymbol{q}_h$ and u_h.
2. On each K, compute ν_K and ε_K (Remark 3.2) and then $\tilde{\eta}_K$.
3. Determine $\mathcal{C}_h$ from $\boldsymbol{q}_h \cdot \boldsymbol{n}$.
4. Compute v_h^0 by local averaging, and then $\eta_{C,E}$ and $\mathcal{Z}_h$ according to (7).
5. Assemble η_K and osc_K. The function v_h is used only in the proof.

6 Identity and origin of the oscillation

Lemma 6.1 (identity)**.** *Let $\boldsymbol{\tau} \in H(\mathrm{div}, \Omega)$ with $\boldsymbol{\tau} \cdot \boldsymbol{n} \in L^2(\Gamma_C)$ and $\boldsymbol{\tau} \cdot \boldsymbol{n} \leq 0$ on Γ_C, and let $v \in \mathcal{K}$. With $e := u - v$,*

$$\|\nabla e\|^2 + \|\boldsymbol{q} - \boldsymbol{\tau}\|^2 + 2\langle -\boldsymbol{\tau} \cdot \boldsymbol{n}, u \rangle_{\Gamma_C} + 2\langle -\boldsymbol{q} \cdot \boldsymbol{n}, v \rangle_{\Gamma_C} = \|\boldsymbol{\tau} + \nabla v\|^2 + 2\langle -\boldsymbol{\tau} \cdot \boldsymbol{n}, v \rangle_{\Gamma_C} + 2\,(f - \operatorname{div} \boldsymbol{\tau},\, e). \tag{10}$$

The four terms on the left are ≥ 0.

Proof. Step 1 (algebra). Since $\boldsymbol{q} = -\nabla u$, we have $\boldsymbol{\tau} + \nabla v = (\boldsymbol{\tau} - \boldsymbol{q}) - \nabla e$. Therefore

$$\|\boldsymbol{\tau} + \nabla v\|^2 = \|\boldsymbol{q} - \boldsymbol{\tau}\|^2 + \|\nabla e\|^2 + 2(\boldsymbol{q} - \boldsymbol{\tau}, \nabla e).$$

Step 2 (integration by parts). Since $e \in H_D^1$, (3) applied to $\boldsymbol{q} - \boldsymbol{\tau}$ gives

$$(\boldsymbol{q} - \boldsymbol{\tau}, \nabla e) = \langle (\boldsymbol{q} - \boldsymbol{\tau}) \cdot \boldsymbol{n}, e \rangle_{\Gamma_C} - (\mathrm{div}(\boldsymbol{q} - \boldsymbol{\tau}), e) = \langle (\boldsymbol{q} - \boldsymbol{\tau}) \cdot \boldsymbol{n}, e \rangle_{\Gamma_C} - (f - \operatorname{div} \boldsymbol{\tau}, e).$$

Step 3 (boundary). By Lemma 1.1, $\langle \boldsymbol{q} \cdot \boldsymbol{n}, u \rangle_{\Gamma_C} = 0$, and then

$$\langle (\boldsymbol{q} - \boldsymbol{\tau}) \cdot \boldsymbol{n}, u - v \rangle_{\Gamma_C} = \langle -\boldsymbol{\tau} \cdot \boldsymbol{n}, u \rangle_{\Gamma_C} + \langle -\boldsymbol{q} \cdot \boldsymbol{n}, v \rangle_{\Gamma_C} - \langle -\boldsymbol{\tau} \cdot \boldsymbol{n}, v \rangle_{\Gamma_C}.$$

Rearranging gives (10). As for the signs: $-\boldsymbol{\tau} \cdot \boldsymbol{n} \geq 0$ and $u \geq 0$ on Γ_C, and $\langle -\boldsymbol{q} \cdot \boldsymbol{n}, v \rangle_{\Gamma_C} \geq 0$ by Lemma 1.1. □

Remark 6.2 (where the oscillation comes from, and its order)**.** In the proof, f enters in only one place, Step 2. Integrating $(\boldsymbol{q}-\boldsymbol{\tau}, \nabla e)$ by parts produces $\operatorname{div}(\boldsymbol{q}-\boldsymbol{\tau}) = f - \operatorname{div}\boldsymbol{\tau}$, which is the residual of the equilibrium equation $\operatorname{div}\boldsymbol{q} = f$ evaluated at $\boldsymbol{\tau}$. For $\boldsymbol{\tau} = \boldsymbol{q}_h$, the mixed method satisfies this equation exactly up to projection: $\operatorname{div}\boldsymbol{q}_h = Q_h f$, by (H1). Thus the residual $f - \operatorname{div}\boldsymbol{q}_h = f - Q_h f$ has no discrete part, and there is no term computable from $\boldsymbol{q}_h$ to add to the indicator. For the same reason, the estimator (25) of MRV has no equilibrium term either. What remains is pure data. Since Q_h reproduces constants on each K, the Payne–Weinberger inequality [11] on convex elements gives

$$(f - Q_h f, e) = \sum_K (f - Q_h f,\, e - \bar{e}_K)_K \le \sum_K \|f - Q_h f\|_K \frac{h_K}{\pi} \|\nabla e\|_K \le \operatorname{osc} \|\nabla e\|.$$

Order (corrected with respect to version 1). If f is smooth, $\operatorname{osc} = O(h^{m+2})$. For RT_0 ($m=0$, $k=1$) this is of higher order than the error, which is $O(h)$. For BDM_p ($m = p-1$, $k = p+1$) it is $O(h^{p+1})$, *of the same order* as $\|\boldsymbol{q} - \boldsymbol{q}_h\|$ and $\|\nabla_h(u - \nu_h)\|$. In that case osc is a computable data term, as in Lovadina–Stenberg [9], which is added to the estimator for marking and for error control; it vanishes if $f \in \mathbb{P}^m(\mathcal{T}_h)$. The analogue in MRV is $\operatorname{osc}(\boldsymbol{q})$ of [10, (23)], which appears in [10, (22), (26)] through the Fortin operator (Assumption 3.4); [10, Rem. 3.9] relates $\operatorname{div}(\boldsymbol{q} - \boldsymbol{q}_h) = f - Q_h f$ to a data oscillation. The difference is that $\operatorname{osc}(\boldsymbol{q})$ involves the exact flux and is not computable, whereas (8) depends only on f. Table 2 illustrates this: with BDM_1 and $f \notin \mathbb{P}^0$, osc is comparable to the error.

Remark 6.3 (it cannot be dropped)**.** If $f \notin \mathbb{P}^m(\mathcal{T}_h)$, some data term is unavoidable. If $\eta = 0$, then $\boldsymbol{q}_h = -\nabla\nu_h$, $\nu_h \in H^1_D$ and $v_h^0 = \nu_h$. Moreover $\nu_h \ge 0$ and $\partial_n \nu_h = -\boldsymbol{q}_h \cdot \boldsymbol{n} \ge 0$ on Γ_C. The complementarity $\nu_h\, \partial_n \nu_h = 0$ also holds: on $\mathcal{Z}_h$ because $\nu_h = 0$, on $\mathcal{C}_h \setminus \mathcal{Z}_h$ because $(\nu_h, -\boldsymbol{q}_h \cdot \boldsymbol{n})_E = 0$, and on $\mathcal{F}_h^C \setminus \mathcal{C}_h$ because $\boldsymbol{q}_h \cdot \boldsymbol{n} \equiv 0$. Since $-\Delta\nu_h = Q_h f$, ν_h is the solution of (1) with data $Q_h f$. The error is then $\|\nabla(u(f) - u(Q_h f))\| \le \|f - Q_h f\|_{H_D^{-1}} \le \operatorname{osc}$, which is positive in general.

7 Reliability

As in [10, (26)–(27)], the error of the postprocessing is measured in the broken norm with jumps

$$\|w\|_{1,h}^2 := \|\nabla_h w\|^2 + \sum_{F \in \mathcal{F}_h^i} h_F^{-1} \|[w]\|_F^2 + \sum_{F \in \mathcal{F}_h^D} h_F^{-1} \|w\|_F^2.$$

Since $u \in H^1_D$, the jumps of $u - \nu_h$ are those of ν_h, and $u - \nu_h = -\nu_h$ on Γ_D.

Theorem A. *Assume (H1) and (H2). Let*

$$A^2 := \|\nabla(u - v_h)\|^2 + \|\boldsymbol{q} - \boldsymbol{q}_h\|^2 + 2\langle -\boldsymbol{q}_h \cdot \boldsymbol{n}, u\rangle_{\Gamma_C} + 2\langle -\boldsymbol{q} \cdot \boldsymbol{n}, v_h\rangle_{\Gamma_C},$$
$$\eta_{\mathrm{rec}}^2 := \|\boldsymbol{q}_h + \nabla v_h\|^2 + 2(v_h, -\boldsymbol{q}_h \cdot \boldsymbol{n})_{\Gamma_C}.$$

(a) $A \le \eta_{\mathrm{rec}} + 2\operatorname{osc}$ *and* $\eta_{\mathrm{rec}} \le A + \operatorname{osc}$.

(b) $\|\boldsymbol{q} - \boldsymbol{q}_h\| + \|u - \nu_h\|_{1,h} \le 2\|\boldsymbol{q}_h + \nabla_h \nu_h\| + (3C_{\mathcal{A}} + 1)J + 2S + 4\operatorname{osc} \le C_{\mathrm{rel}}\, \eta + 4\operatorname{osc}$, *with* $C_{\mathrm{rel}} := \big(8 + (3C_{\mathcal{A}} + 1)^2\big)^{1/2}$.

Neither saturation nor a Fortin operator is used.

Proof. (a) By Lemma 4.3(b), $v_h \in \mathcal{K}$. Lemma 6.1 with $\boldsymbol{\tau} = \boldsymbol{q}_h$ and $v = v_h$ gives $A^2 = \eta_{\mathrm{rec}}^2 + 2(f - Q_h f, u - v_h)$. By Remark 6.2, $|2(f - Q_h f, u - v_h)| \le 2\operatorname{osc} A$. Then $A^2 \le \eta_{\mathrm{rec}}^2 + 2\operatorname{osc} A$, whence $A \le \operatorname{osc} + (\operatorname{osc}^2 + \eta_{\mathrm{rec}}^2)^{1/2} \le \eta_{\mathrm{rec}} + 2\operatorname{osc}$. Likewise, $\eta_{\mathrm{rec}}^2 \le A^2 + 2\operatorname{osc} A \le (A + \operatorname{osc})^2$.

(b) By Lemma 4.3(c) and (d), $\eta_{\mathrm{rec}} \le \|\boldsymbol{q}_h + \nabla_h \nu_h\| + C_{\mathcal{A}} J + S$. Moreover $\|\boldsymbol{q} - \boldsymbol{q}_h\| \le A$ and

$$\|u - \nu_h\|_{1,h} \le \|\nabla_h(u - \nu_h)\| + J \le A + (C_{\mathcal{A}} + 1)J.$$

Using (a) gives the first inequality. The second follows from Cauchy–Schwarz and (9). □

Remark 7.1. The version with $\eta_{\mathrm{rec},K} := \|\boldsymbol{q}_h + \nabla v_h\|_K$, plus the complementarity on $\mathcal{C}_h \setminus \mathcal{Z}_h$, is computable and two-sided with constant 1 (part (a)). Locally, $\eta_{\mathrm{rec},K} \le \|\boldsymbol{q} - \boldsymbol{q}_h\|_K + \|\nabla(u - v_h)\|_K$, because $\boldsymbol{q}_h + \nabla v_h = (\boldsymbol{q}_h - \boldsymbol{q}) - \nabla(u - v_h)$. The price is that it measures the error of v_h rather than that of ν_h. The indicator η_K is its localized upper bound, with the structure of (25) in MRV. If $u_D \neq 0$, one imposes $v_h = \mathcal{I} u_D$ on Γ_D and a data term in u_D appears, as in MRV.

Remark 7.2 (no trace term on Γ_C)**.** On Γ_D, the term $h_F^{-1}\|u_D - \nu_h\|_F^2$ of the norm involves only data and is literally an estimator term. On Γ_C, u is unknown except on the actual contact set $\Gamma_C^0 := \{u = 0\}$, and the norm carries no trace term because of a global, not a local, obstruction:

- $\sum_{E \subset \Gamma_C} h_E^{-1}\|u - \nu_h\|_E^2$ is not controlled by energy quantities: a smooth shift of height c costs $O(c)$ in energy and c^2/h in that term.
- $\sum_{E \subset \Gamma_C^0} h_E^{-1}\|\nu_h\|_E^2$ is trivially efficient but not reliable. Let u have strict contact ($\partial_n u \ge g > 0$) on a part of Γ_C, $\boldsymbol{q}_h$ the canonical interpolant of $\boldsymbol{q}$ and ν_h the Bernstein interpolant of $u + c\varphi$, with φ smooth, equal to 1 on that part and zero near Γ_D; (H1) and (H2) hold. On each edge the trace term is c^2 and the complementarity term $\approx 2cgh$, so for $c > 2gh$ the minimum rule picks complementarity. Summing over the $\sim 1/h$ edges, $\eta^2 \sim c^2 + c$ while the term is $\sim c^2/h$: the ratio $\sim c/h$ is unbounded.

The failure requires a trace error larger than h; asymptotically the rule picks the trace on all edges of actual contact except the transition ones. In E1, E2 and E3 with $k \le 2$, adding that term to $\mathbb{E}$ does not change $\eta/\mathbb{E}$ in the second digit: it is at most 5 % of $\mathbb{E}$ and lives entirely on edges of $\mathcal{Z}_h$.

8 Local efficiency

Let $\mathbb{E}_K$ be the restriction to K of the error in the MRV norm:

$$\mathbb{E}_K^2 := \|\boldsymbol{q} - \boldsymbol{q}_h\|_K^2 + \|\nabla(u - \nu_K)\|_K^2 + \tfrac{1}{2}\sum_{F \in \mathcal{F}_K^i} h_F^{-1}\|[\nu_h]\|_F^2 + \sum_{F \in \mathcal{F}_K^D} h_F^{-1}\|\nu_h\|_F^2,$$

so that $\sum_K \mathbb{E}_K^2 = \|\boldsymbol{q} - \boldsymbol{q}_h\|^2 + \|u - \nu_h\|_{1,h}^2 =: \mathbb{E}^2$.

Proposition 8.1. *(i)* $\tilde{\eta}_K \le \|\boldsymbol{q}_h + \nabla\nu_K\|_K \le \|\boldsymbol{q} - \boldsymbol{q}_h\|_K + \|\nabla(u - \nu_K)\|_K$.

(ii) The jump and Dirichlet terms of η_K^2 *are summands of* $\mathbb{E}_K^2$ *(constant* 1*). In particular, if* K *has no edge in* $\mathcal{C}_h$*, then* $\eta_K \le 2\,\mathbb{E}_K$.

(iii) If $E \in \mathcal{F}_K^C \cap \mathcal{C}_h$ *and* $u|_E = 0$*, then* $\eta_{C,E} \le h_E^{-1/2}\|\nu_h\|_E \le C(\|\nabla(u - \nu_K)\|_K + h_K^{-1}\|\Pi_0(u - u_h)\|_K)$.

(iv) If $E \in \mathcal{F}_K^C \cap \mathcal{C}_h$ *and* $\boldsymbol{q}\cdot\boldsymbol{n} \in L^2(E)$*, let* $E_0 := E \cap \{u = 0\}$ *and* $E_+ := E \setminus E_0$*. Then*

$$\eta_{C,E}^2 \le 2\|v_h^0 - u\|_{E_0}(\|\boldsymbol{q}\cdot\boldsymbol{n}\|_{E_0} + \|(\boldsymbol{q} - \boldsymbol{q}_h)\cdot\boldsymbol{n}\|_{E_0}) + 2\|(\boldsymbol{q} - \boldsymbol{q}_h)\cdot\boldsymbol{n}\|_{E_+}(\|u\|_{E_+} + \|v_h^0 - u\|_{E_+}).$$

The constant C *depends only on shape regularity and on* k.

Proof. (i) By Lemma 3.1(c) and because $\boldsymbol{q}_h + \nabla\nu_K = (\boldsymbol{q}_h - \boldsymbol{q}) - \nabla(u - \nu_K)$.

(ii) Immediate. With (i), $\tilde{\eta}_K^2 + \|\boldsymbol{q}_h + \nabla\nu_K\|_K^2 \le 2(\|\boldsymbol{q} - \boldsymbol{q}_h\|_K + \|\nabla(u - \nu_K)\|_K)^2 \le 4(\|\boldsymbol{q} - \boldsymbol{q}_h\|_K^2 + \|\nabla(u - \nu_K)\|_K^2)$.

(iii) Let $w := u - \nu_K$; on E we have $\nu_h = -w$. The trace inequality gives $h_E^{-1/2}\|w\|_E \le C(h_K^{-1}\|w\|_K + \|\nabla w\|_K)$. By Poincaré, $h_K^{-1}\|w\|_K \le \pi^{-1}\|\nabla w\|_K + h_K^{-1}\|\Pi_0 w\|_K$, and by (H2), $\Pi_0 w = \Pi_0(u - u_h)$. Moreover, $\eta_{C,E}^2 \le h_E^{-1}\|\nu_h\|_E^2$ by the definition of the minimum.

(iv) The function $u\,\boldsymbol{q}\cdot\boldsymbol{n}$ is ≤ 0 on Γ_C and has zero integral by Lemma 1.1, so it vanishes almost everywhere; in particular, $\boldsymbol{q}\cdot\boldsymbol{n} = 0$ on E_+. Since $u = 0$ on E_0,

$$\eta_{C,E}^2 \le 2(v_h^0, -\boldsymbol{q}_h\cdot\boldsymbol{n})_E = 2(v_h^0 - u, -\boldsymbol{q}_h\cdot\boldsymbol{n})_{E_0} + 2(v_h^0, (\boldsymbol{q} - \boldsymbol{q}_h)\cdot\boldsymbol{n})_{E_+},$$

and one uses $|\boldsymbol{q}_h\cdot\boldsymbol{n}| \le |\boldsymbol{q}\cdot\boldsymbol{n}| + |(\boldsymbol{q} - \boldsymbol{q}_h)\cdot\boldsymbol{n}|$, $v_h^0 \le u + |v_h^0 - u|$ and Cauchy–Schwarz. □

Remark 8.2 (transition edges)**.** By (iii), $\eta_{C,E}$ is locally efficient on edges of actual contact; the term $h_K^{-1}\|\Pi_0(u-u_h)\|_K$ appears only there. It lacks a local bound only on the edges of $\mathcal{C}_h$ where $u \not\equiv 0$. Under (H3), and if the discrete active set converges to the continuous one (Remark 2.2), these edges are only those adjacent to transition points. With finitely many transitions, as in hypothesis (A) of Christof–Haubner [3], there are $O(1)$ such edges. In (iv), the cross terms are $\|v_h^0 - u\|_{E_0}\|\boldsymbol{q}\cdot\boldsymbol{n}\|_{E_0}$ and $\|(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}\|_{E_+}\|u\|_{E_+}$: an error multiplied by the flux on the contact part of the edge or by the trace on its separation part, both of which vanish at the transition point. If the edge contains no transition, one of the two summands disappears. For an $r^{3/2}$-type transition, the *order count* gives h^3 for both, the same as the squared local error. This is not a bound.

Remark 8.3 (comparison with the primal case)**.** For Signorini problems in linear elasticity with P_1 elements, Krause–Veeser–Walloth [8] prove local efficiency of the volume terms; that of the contact-stress residual only at nodes that are not full-contact nodes, that of the complementarity term under assumptions on the semi-contact nodes, and they give no lower bounds for the terms that appear with a non-discrete gap. Partial efficiency at the free boundary is thus the usual situation, and the one in this note is comparable.

9 Reaction and Signorini conditions on the whole boundary

Now let $-\Delta u + cu = f$ with constant $c > 0$, and allow $\Gamma_D = \emptyset$ (the FHS examples [4] have $c = 1$ and Signorini conditions on all of $\partial\Omega$). Lemma 1.1 holds with the form $(\nabla u, \nabla v) + c(u,v)$, and (3) with $w \in H^1(\Omega)$ if $\Gamma_D = \emptyset$. The mixed method gives $\operatorname{div}\boldsymbol{q}_h + cu_h = Q_h f$.

Lemma 9.1 (identity with reaction)**.** *Let $\boldsymbol{\tau}$ and v be as in Lemma 6.1, $e := u - v$ and $r := f - cv - \operatorname{div}\boldsymbol{\tau}$. Then*

$$\|\nabla e\|^2 + 2c\|e\|^2 + \|\boldsymbol{q}-\boldsymbol{\tau}\|^2 + 2\langle -\boldsymbol{\tau}\cdot\boldsymbol{n}, u\rangle_{\Gamma_C} + 2\langle -\boldsymbol{q}\cdot\boldsymbol{n}, v\rangle_{\Gamma_C} = \|\boldsymbol{\tau} + \nabla v\|^2 + 2\langle -\boldsymbol{\tau}\cdot\boldsymbol{n}, v\rangle_{\Gamma_C} + 2(r,e).$$

Proof. This is the proof of Lemma 6.1, with $\operatorname{div}(\boldsymbol{q}-\boldsymbol{\tau}) = f - cu - \operatorname{div}\boldsymbol{\tau} = r - ce$ in Step 2. □

Theorem B. *Let $m = 0$ (RT_0 or BDM_1) and assume (H1) and (H2). Let*

$$A_c^2 := \|\nabla(u-v_h)\|^2 + c\|u-v_h\|^2 + \|\boldsymbol{q}-\boldsymbol{q}_h\|^2 + 2\langle -\boldsymbol{q}_h\cdot\boldsymbol{n}, u\rangle_{\Gamma_C} + 2\langle -\boldsymbol{q}\cdot\boldsymbol{n}, v_h\rangle_{\Gamma_C},$$
$$\operatorname{osc}_c^2 := \sum_K \Big(\frac{h_K}{\pi}\Big)^2 \|(I-\Pi_0)(f-cv_h)\|_K^2.$$

Then $A_c \le \big(\eta_{\mathrm{rec}}^2 + c\|\Pi_0(\nu_h - v_h)\|^2\big)^{1/2} + 2\operatorname{osc}_c$, and $c\|\Pi_0(\nu_h - v_h)\|^2 \le c\,C_{\mathcal{A}}^2 h^2 J^2$.

Proof. With $\boldsymbol{\tau} = \boldsymbol{q}_h$ and $\Pi_0\nu_h = u_h$ we have $r = f - cv_h - \Pi_0 f + c\,\Pi_0\nu_h = (I-\Pi_0)(f-cv_h) + c\,\Pi_0(\nu_h - v_h)$. By Payne–Weinberger and Young,

$$2(r,e) \le 2\operatorname{osc}_c\|\nabla e\| + c\|\Pi_0(\nu_h - v_h)\|^2 + c\|e\|^2.$$

The last term is absorbed into $2c\|e\|^2$ of Lemma 9.1, and one concludes as in Theorem A(a). The final bound is Lemma 4.3(d). □

Remark 9.2. (1) osc_c is computable, but it depends on v_h and is of the same order as the error; for $c = 0$ it reduces to osc. (2) For $m \ge 1$, the MRV constraint only fixes the mean, and the term $c((Q_h - \Pi_0)(u_h - \nu_h), e)$ remains, of lower order than the error. It is avoided by imposing Stenberg's original constraint $Q_h\nu_h = u_h$ in the postprocessing: then $r = (I - Q_h)(f - cv_h) + c\,Q_h(\nu_h - v_h)$ and Theorem B holds with Q_h in place of Π_0. This is what is done in Section 10.1 for FHS 4.2 with BDM_2. (3) The bounds are not robust in c; the singularly perturbed regime of FHS 4.4 is not covered. (4) The experiments with FHS 4.2 use the same η of Definition 5.1, without the two terms of this theorem.

10 Numerical experiments

Implementation. Python prototype (NumPy/SciPy), with BDM_1 ($k = 2$) and RT_0 ($k = 1$) on *criss-cross* meshes refined by newest-vertex bisection. Uniform refinement is two bisections per level, so h is halved. The variational inequality (4) is solved with the primal-dual active set method of Remark 2.3; for BDM_1 the constraints are the endpoint values of $\boldsymbol{q}_h \cdot \boldsymbol{n}$, which is the exact pointwise condition. The constrained postprocessing uses `lsq_linear/bvls`, and $\tilde{\eta}_K$ is computed with $V_K = \mathbb{P}_*^{k+1}(K)$. For general BDM_p (Section 10.1), the edge degrees of freedom are the degree-p Bernstein coefficients of $\boldsymbol{q}_h \cdot \boldsymbol{n}$, so the constraints are bounds on degrees of freedom (for $p \geq 2$, only sufficient ones); the interior ones are moments against $\nabla\mathbb{P}^{p-1}$ and $\operatorname{curl}(b_K\mathbb{P}^{p-2})$, and the local QP of the postprocessing is solved with a dense primal-dual active set method. Three versions of the contact term are compared: $\eta_{\min}$ (the minimum rule (7)), η_{FHS} (complementarity only, $2(v_h^0, -\boldsymbol{q}_h \cdot \boldsymbol{n})_E$ on all of $\mathcal{C}_h$) and η_{tr} (trace only, $h_E^{-1}\|\nu_h\|_E^2$ on all of $\mathcal{C}_h$). The error is $\mathbb{E} = (\|\boldsymbol{q} - \boldsymbol{q}_h\|^2 + \|u - \nu_h\|_{1,h}^2)^{1/2}$ and the effectivity is $\eta/\mathbb{E}$. N is the number of degrees of freedom of the mixed problem. The adaptive runs use Dörfler marking ($\theta = 0.5$) on $\eta_{\min,K}^2 + \operatorname{osc}_K^2$. The column "mis" gives the number of edges of $\mathcal{C}_h$ where $u \not\equiv 0$ and of edges outside $\mathcal{C}_h$ where $\partial_n u \not\equiv 0$.

Examples.

- E0 Validation without contact on $(-1,1) \times (0,1)$, Dirichlet on the whole boundary: $u = \sin(\pi x)\sin(\pi y) + xy$ ($f \notin \mathbb{P}^0$) and $u = e^x \sin y - x^2/2$ ($f = 1$).
- E1 $u = -r^{7/2}\sin(7\theta/2)$ centred at $(x_0, 0)$, $\Gamma_C = (-1,1) \times \{0\}$, exact Dirichlet data elsewhere, $f = 0$. Contact for $x > x_0$, separation for $x < x_0$, $u \in H^{9/2-\varepsilon}$. With $x_0 = \frac{1}{3}$ the transition is never a node; with $x_0 = 0$ it always is.
- E2 As E1 with $r^{3/2}\sin(3\theta/2)$ and $x_0 = \frac{1}{3}$; $u \in H^{5/2-\varepsilon}$.
- E3 FHS 4.2 [4]: $-\Delta u + u = f$ on $(-1,1)\times(0,1)$ with Signorini conditions on the whole boundary and a piecewise polynomial solution; mesh aligned with $x = 0$, transitions at nodes.
- E4 BGK 7.3 [1]: $(-1,1)^2$, $f = -1$, obstacle $\chi = \min\{\frac{1}{2}(|x_1| - \frac{1}{2}), 0\}$ at the bottom, Dirichlet at the top and on $\{1\} \times (0,1)$, Neumann elsewhere (Remark 1.2). No exact solution.
- E5 Effect of the Bernstein constraint: E1 ($x_0 = \frac{1}{3}$) and E3 with the unconstrained Stenberg postprocessing ν_h^0.

$\#\mathcal{T}_h$	$\mathbb{E}$	$\eta/\mathbb{E}$	$\operatorname{osc}/\mathbb{E}$	$(\eta^2 + \operatorname{osc}^2)^{1/2}/\mathbb{E}$	$\mathbb{E}$	$\eta/\mathbb{E}$
8	2.17e+0	0.22	1.20	1.22	1.15e-1	0.76
32	7.19e-1	0.15	1.11	1.12	2.99e-2	0.78
128	1.83e-1	0.17	1.12	1.13	7.56e-3	0.79
512	4.59e-2	0.18	1.12	1.13	1.90e-3	0.80
2048	1.15e-2	0.18	1.12	1.13	4.75e-4	0.80
8192	2.87e-3	0.18	1.12	1.13	1.19e-4	0.80

Table 2: E0, BDM_1. Left: $f \notin \mathbb{P}^0$; right: $f = 1 \in \mathbb{P}^0$.

$\#\mathcal{T}_h$	$\|\boldsymbol{q} - \boldsymbol{q}_h\|$	r	$\|\nabla_h(u - \nu_h)\|$	r	$\|u - \nu_h\|$	r	$\|\Pi_0 u - u_h\|$	r	$\frac{\eta_{\min}}{\mathbb{E}}$	$\frac{\eta_{\mathrm{FHS}}}{\mathbb{E}}$	$\frac{\eta_{\mathrm{tr}}}{\mathbb{E}}$	mis
8	6.93e-1	–	6.80e-1	–	7.71e-2	–	4.40e-2	–	0.80	0.82	0.80	1/0
32	1.75e-1	1.99	1.61e-1	2.07	9.34e-3	3.04	5.80e-3	2.92	0.79	0.84	0.79	0/1
128	4.36e-2	2.00	4.02e-2	2.00	1.11e-3	3.07	6.42e-4	3.18	0.80	0.86	0.80	1/0
512	1.09e-2	2.00	1.01e-2	2.00	1.39e-4	3.00	8.07e-5	2.99	0.80	0.91	0.80	0/1
2048	2.73e-3	2.00	2.51e-3	2.00	1.74e-5	3.00	1.00e-5	3.01	0.80	1.01	0.80	1/0
8192	6.81e-4	2.00	6.28e-4	2.00	2.17e-6	3.00	1.25e-6	3.00	0.80	1.19	0.80	0/1
32768	1.70e-4	2.00	1.57e-4	2.00	2.72e-7	3.00	1.57e-7	3.00	0.80	1.47	0.80	1/0

Table 3: E1, $r^{7/2}$, $x_0 = \frac{1}{3}$ (never a node), BDM_1 ($k = 2$). r: rate with respect to h.

$\#\mathcal{T}_h$	$x_0=\frac{1}{3}$, $k=2$		$x_0=\frac{1}{3}$, $k=1$		$x_0=0$, $k=2$		$x_0=0$, $k=1$	
	min	FHS	min	FHS	min	FHS	min	FHS
8	0.80	0.82	0.48	0.48	0.82	0.93	0.48	0.51
32	0.79	0.84	0.54	0.54	0.81	0.96	0.55	0.56
128	0.80	0.86	0.56	0.56	0.81	1.08	0.57	0.57
512	0.80	0.91	0.57	0.57	0.81	1.30	0.57	0.58
2048	0.80	1.01	0.57	0.57	0.81	1.64	0.58	0.58
8192	0.80	1.19	0.58	0.58	0.80	2.18	0.58	0.58
32768	0.80	1.47	0.58	0.58	0.80	2.97	0.58	0.58

Table 4: E1: effectivities of $\eta_{\min}$ and η_{FHS} for $k=2$ (BDM_1) and $k=1$ (RT_0).

$\#\mathcal{T}_h$	$\|\boldsymbol{q}-\boldsymbol{q}_h\|$	r	$\|\nabla_h(u-\nu_h)\|$	r	$\frac{\eta_{\min}}{\mathbb{E}}$	$\frac{\eta_{\mathrm{FHS}}}{\mathbb{E}}$	$\frac{\eta_{\mathrm{tr}}}{\mathbb{E}}$	$\#(\mathcal{C}_h\setminus\mathcal{Z}_h)$	mis
8	7.81e-2	–	9.79e-2	–	0.98	1.54	0.98	0	1/0
32	4.37e-2	0.84	4.70e-2	1.06	1.28	1.28	1.40	1	1/0
128	1.06e-2	2.05	1.25e-2	1.91	0.96	1.50	0.96	0	1/0
512	5.46e-3	0.95	5.94e-3	1.07	1.28	1.28	1.39	1	1/0
2048	1.34e-3	2.03	1.57e-3	1.92	0.96	1.50	0.96	0	1/0
8192	6.82e-4	0.97	7.45e-4	1.07	1.28	1.28	1.39	1	1/0
32768	1.68e-4	2.02	1.96e-4	1.93	0.96	1.50	0.96	0	1/0

Table 5: E2, $r^{3/2}$, $x_0=\frac{1}{3}$, BDM_1, uniform refinement. r: rate with respect to h.

$k=2$ (BDM_1)			$k=1$ (RT_0)		
N	$\mathbb{E}$	$\eta/\mathbb{E}$	N	$\mathbb{E}$	$\eta/\mathbb{E}$
38	1.37e-1	0.98	23	7.68e-1	0.83
116	2.55e-2	0.87	58	4.93e-1	0.64
261	1.13e-2	0.78	216	2.65e-1	0.59
924	3.10e-3	0.73	549	1.50e-1	0.58
2277	1.26e-3	0.70	1845	8.04e-2	0.58
7352	3.86e-4	0.69	4631	5.34e-2	0.58
23126	1.22e-4	0.68	14287	3.09e-2	0.58
54050	5.21e-5	0.68	42851	1.72e-2	0.58
164529	1.71e-5	0.68	128052	9.52e-3	0.58
rate $N^{-1.00}$			rate $N^{-0.51}$		

Table 6: E2, adaptive refinement (selected iterations). Rates fitted over the second half of the iterations.

$\#\mathcal{T}_h$	$\|\boldsymbol{q}-\boldsymbol{q}_h\|$	r	$\|\nabla_h(u-\nu_h)\|$	r	$\frac{\eta_{\min}}{\mathbb{E}}$	$\frac{\eta_{\mathrm{FHS}}}{\mathbb{E}}$	$\frac{\mathrm{osc}}{\mathbb{E}}$	$\frac{(\eta^2+\mathrm{osc}^2)^{1/2}}{\mathbb{E}}$	$\frac{\eta_{\min}}{\mathbb{E}}$	$\frac{\mathrm{osc}}{\mathbb{E}}$
8	7.18e-1	–	7.15e-1	–	1.06	1.06	1.23	1.62	1.10	0.74
32	2.40e-1	1.58	2.36e-1	1.60	0.93	0.99	1.19	1.51	1.06	0.33
128	6.33e-2	1.92	6.10e-2	1.95	0.77	0.83	1.15	1.39	0.98	0.15
512	1.60e-2	1.98	1.53e-2	1.99	0.67	0.74	1.13	1.32	0.88	0.07
2048	4.01e-3	1.99	3.84e-3	2.00	0.61	0.72	1.13	1.28	0.82	0.04
8192	1.01e-3	2.00	9.61e-4	2.00	0.58	0.79	1.12	1.26	0.78	0.02
32768	2.52e-4	2.00	2.40e-4	2.00	0.57	0.95	1.12	1.26	0.76	0.01

Table 7: E3, FHS 4.2. Columns 2–10: BDM_1; last two: RT_0. r: rate with respect to h.

$\#\mathcal{T}_h$	$r^{7/2}$, $x_0 = \frac{1}{3}$				FHS 4.2			
	δ_∇	δ_{L^2}	n_{act}	$\min(\nu_h^0 - \chi)$	δ_∇	δ_{L^2}	n_{act}	$\min(\nu_h^0 - \chi)$
8	3.59e-2	2.55e-2	2	-1.39e-1	1.97e-2	1.21e-3	5	-1.70e-1
32	7.94e-3	1.30e-2	2	-2.83e-2	7.77e-3	4.68e-3	8	-3.10e-2
128	1.71e-3	2.20e-3	4	-2.58e-3	8.84e-3	7.89e-4	12	-7.29e-3
512	1.23e-3	1.50e-3	5	-3.41e-4	2.58e-3	6.82e-5	26	-1.15e-3
2048	5.68e-4	6.59e-4	12	-4.37e-5	9.24e-4	1.74e-6	50	-1.47e-4
8192	2.89e-4	3.35e-4	21	-5.53e-6	1.47e-3	4.11e-7	98	-1.67e-5
32768	1.44e-4	1.66e-4	44	-6.95e-7	1.06e-3	9.59e-8	194	-2.29e-6

Table 8: E5, BDM_1: relative difference between ν_h (constrained) and ν_h^0 (unconstrained Stenberg) in $\|\nabla_h(u - \cdot)\|$ (δ_∇) and $\|u - \cdot\|$ (δ_{L^2}); n_{act} = elements where the constraint is active; $\min(\nu_h^0 - \chi)$ on Γ_C.

uniform				adaptive			
$k = 2$		$k = 1$		$k = 2$		$k = 1$	
N	η	N	η	N	η	N	η
272	9.03e-2	168	3.61e-1	272	9.03e-2	168	3.61e-1
1056	5.70e-2	656	2.01e-1	788	2.26e-2	467	2.04e-1
4160	3.78e-2	2592	1.15e-1	2223	8.82e-3	1601	1.10e-1
16512	2.37e-2	10304	6.52e-2	7551	2.61e-3	5011	6.06e-2
65792	1.66e-2	41088	4.05e-2	21142	9.24e-4	11804	3.88e-2
				68727	2.82e-4	36661	2.25e-2
				188735	1.02e-4	111351	1.29e-2
$N^{-0.30}$		$N^{-0.38}$		$N^{-1.01}$		$N^{-0.51}$	

Table 9: E4, BGK 7.3: estimator $\eta_{\min}$; fitted rates in the last row.

Observations.

(1) *Oscillation (E0, Table 2).* With BDM_1 and $f \notin \mathbb{P}^0$, $\eta/\mathbb{E} \approx 0.18$ but $\text{osc}/\mathbb{E} \approx 1.12$: the error that the indicator does not see is explained by the oscillation, which is of the same order, and $(\eta^2 + \text{osc}^2)^{1/2}/\mathbb{E} \approx 1.13$ is stable. With $f \in \mathbb{P}^0$, $\eta/\mathbb{E} \to 0.80$. This confirms the correction in Remark 6.2.

(2) *Superconvergence with contact (E1, Table 3).* With the transition away from the nodes and $k = 2$, the rates are optimal: 2 for $\boldsymbol{q}_h$ and $\nabla_h \nu_h$, and 3 for ν_h in L^2 and for $\Pi_0 u - u_h$. The contact constraint does not destroy superconvergence when the transition is smooth; this is the numerical answer to the question that decided whether the project was worth pursuing. With BDM_1, the L^2 rate of ν_h is in general limited by $\|\Pi_0 u - u_h\| = O(h^2)$, also without contact (it is 2 in E0 and E3); the h^3 in E1 is due to $f = 0$.

(3) *Contact term (Tables 3, 4 and 5).* $\eta_{\min}/\mathbb{E}$ is constant: 0.80 for $k = 2$ and 0.58 for $k = 1$. For $k = 1$, η_{FHS} and $\eta_{\min}$ coincide, as the order count predicts. For $k = 2$, $\eta_{\text{FHS}}/\mathbb{E}$ grows from 0.82 to 1.47 with $x_0 = \frac{1}{3}$ and from 0.93 to 2.97 with $x_0 = 0$; the factor per level approaches $\sqrt{2}$, that is, $h^{-1/2}$. In E2, at the levels where the minimum rule chooses complementarity on the transition edge ($\#(\mathcal{C}_h \setminus \mathcal{Z}_h) = 1$), it gives 1.28, against 1.39–1.40 for η_{tr}.

(4) *Singularity and adaptivity (E2, Tables 5 and 6).* On uniform meshes the rates alternate around $h^{3/2}$, the limit imposed by $u \in H^{5/2-\varepsilon}$, and the effectivity alternates between 0.96 and 1.28 without degrading. Adaptive refinement recovers the optimal rates, $N^{-1.00}$ for $k = 2$ and $N^{-0.51}$ for $k = 1$, with stable effectivity (0.68 and 0.58).

(5) *FHS 4.2 (E3, Table 7).* With BDM_1 the rates are 2. Since $f \notin \mathbb{P}^0$, $\text{osc}/\mathbb{E} \approx 1.12$, and $\eta_{\min}/\mathbb{E}$ decreases until it settles near 0.57, while $(\eta^2 + \text{osc}^2)^{1/2}/\mathbb{E} \approx 1.26$ is stable. With RT_0, $\text{osc}/\mathbb{E} \to 0$ and $\eta_{\min}/\mathbb{E} \approx 0.76$. With BDM_1, η_{FHS} starts to grow on the last levels.

(6) *Bernstein constraint (E5, Table 8).* The unconstrained Stenberg postprocessing violates positivity on Γ_C, with $\min(\nu_h^0 - \chi)$ of order 10^{-6} to 10^{-7} on the finest mesh, and the constraint is active on a fixed fraction of the contact edges (44 of 128 in E1, 194 of 384 in E3). But from the third mesh on, the relative difference in the error is below 1 %, and on the finest mesh it is 0.01 % (E1) and 0.1 % (E3). The constraint costs no accuracy in these examples.

(7) *BGK 7.3 (E4, Table 9).* With uniform refinement the estimator decays like $N^{-0.30}$ ($k=2$) and $N^{-0.38}$ ($k=1$); with adaptive refinement, like $N^{-1.01}$ and $N^{-0.51}$, the optimal rates. The contact term is 1.6 % of η on the last adaptive mesh with $k=2$.

(8) *Solver and classification.* The primal-dual active set method terminated in 1 to 20 iterations, with KKT residual $\le 4\cdot 10^{-17}$. The misclassified edges (column "mis") are at most 3 per level and lie within distance $2h$ of the transition.

10.1 Results with BDM_2 ($k=3$), BGK 7.2 and verification of the identity

Additional examples.

E6 BGK 7.2 [1]: $\Omega=(0,1)^2$, $\Gamma_C=(0,1)\times\{0\}$, homogeneous Dirichlet data elsewhere, and $u=-10\,\psi(r)\,r^{3/2}\sin(3\theta/2)$ centred at $(\frac12,0)$. Here ψ is the degree-9 polynomial on $[0,0.45]$ with $\psi(0)=1$, $\psi(0.45)=0$ and derivatives 1 to 4 vanishing at both ends, extended by zero, and $f=10\,\mathrm{Im}(z^{3/2})(\psi''+4\psi'/r)$ with $z=(x-\frac12)+\mathrm{i}y$. In the version of the paper I read, the formula has its exponents flattened; the reading $\sin(3\theta/2)$ is the one consistent with their claim $u\in H^2(\Omega)$, since $(\sin\theta)^{3/2}$ would give $u=-10\psi\,y^{3/2}\notin H^2$. The transition lies at the node $(\frac12,0)$.

ID Verification of the identity: for E3 ($c=1$) and E6 ($c=0$, $u_D=0$) we build the full reconstruction $v_h=\mathcal{A}_{\mathcal{Z}_h}\nu_h$ and evaluate both sides of Lemma 6.1 (or Lemma 9.1) with $\boldsymbol{\tau}=\boldsymbol{q}_h$ and $v=v_h$, using the exact solution. We report the relative defect $|\mathrm{lhs}-\mathrm{rhs}|/\mathrm{lhs}$, the ratio η_{rec}/A, and the empirical ratio $\|\nabla_h(\nu_h-v_h)\|/J$, which is a lower bound for $C_{\mathcal{A}}$.

$\#\mathcal{T}_h$	$\mathbb{E}$	$\eta/\mathbb{E}$	$\mathrm{osc}/\mathbb{E}$	$(\eta^2+\mathrm{osc}^2)^{1/2}/\mathbb{E}$	$\mathbb{E}$	$\eta/\mathbb{E}$
8	1.79e-1	0.00	2.99	2.99	8.42e-3	0.74
32	7.26e-2	0.19	2.12	2.13	1.06e-3	0.77
128	9.66e-3	0.16	2.03	2.04	1.32e-4	0.78
512	1.23e-3	0.15	2.01	2.01	1.65e-5	0.79
2048	1.54e-4	0.15	2.00	2.01	2.07e-6	0.79
8192	1.93e-5	0.15	2.00	2.01	2.59e-7	0.79

Table 10: E0 with BDM_2. Left: $f\notin\mathbb{P}^1$; right: $f=x\in\mathbb{P}^1$.

$\#\mathcal{T}_h$	$\|\boldsymbol{q}-\boldsymbol{q}_h\|$	r	$\|\nabla_h(u-\nu_h)\|$	r	$\|u-\nu_h\|$	r	$\|Q_hu-u_h\|$	r	$\frac{\eta_{\min}}{\mathbb{E}}$	$\frac{\eta_{\mathrm{FHS}}}{\mathbb{E}}$	$\frac{\eta_{\mathrm{tr}}}{\mathbb{E}}$	mis
8	7.07e-2	–	5.41e-2	–	1.13e-2	–	1.09e-2	–	0.87	1.36	0.87	1/0
32	4.88e-3	3.86	7.44e-3	2.86	2.94e-4	5.26	9.60e-5	6.82	1.19	1.35	1.32	1/0
128	7.74e-4	2.66	6.45e-4	3.53	4.70e-5	2.64	4.65e-5	1.05	0.94	1.82	0.94	1/0
512	7.51e-5	3.37	8.35e-5	2.95	7.57e-7	5.96	3.67e-7	6.98	1.07	3.23	1.12	1/0
2048	1.01e-5	2.89	9.02e-6	3.21	1.85e-7	2.03	1.83e-7	1.00	0.97	6.35	0.97	1/0
8192	1.17e-6	3.11	1.14e-6	2.98	2.43e-9	6.25	1.42e-9	7.01	1.01	12.94	1.02	1/0
32768	1.49e-7	2.97	1.37e-7	3.06	7.25e-10	1.74	7.17e-10	0.99	0.98	26.00	0.98	1/0

Table 11: E1, $r^{7/2}$, $x_0=\frac13$, BDM_2 ($k=3$). r: rate with respect to h.

$\#\mathcal{T}_h$	$x_0=\frac13$		$x_0=0$	
	min	FHS	min	FHS
8	0.87	1.36	1.04	1.89
32	1.19	1.35	1.01	2.00
128	0.94	1.82	0.98	2.78
512	1.07	3.23	0.96	5.31
2048	0.97	6.35	0.95	11.06
8192	1.01	12.94	0.94	22.89

N	$\mathbb{E}$	$\eta/\mathbb{E}$
93	9.47e-2	1.41
246	5.80e-3	1.35
687	1.52e-3	0.95
1728	2.59e-4	0.88
5421	4.38e-5	0.86
16296	7.93e-6	0.85
38826	2.12e-6	0.85
110571	4.37e-7	0.85
305268	9.50e-8	0.85

Table 12: Left: E1 with $k=3$, effectivities of $\eta_{\min}$ and η_{FHS}. Right: E2 with $k=3$ and adaptive refinement (selected iterations).

$\#\mathcal{T}_h$	$\|\boldsymbol{q}-\boldsymbol{q}_h\|$	r	$\|\nabla_h(u-\nu_h)\|$	r	$\frac{\eta_{\min}}{\mathbb{E}}$	$\frac{\eta_{\mathrm{FHS}}}{\mathbb{E}}$	$\frac{\eta_{\mathrm{tr}}}{\mathbb{E}}$	$\#(\mathcal{C}_h \setminus \mathcal{Z}_h)$
8	5.17e-2	–	7.27e-2	–	1.41	1.68	1.41	0
32	2.38e-2	1.12	2.23e-2	1.70	1.43	1.48	2.38	1
128	6.43e-3	1.89	9.06e-3	1.30	1.41	1.70	1.41	0
512	2.98e-3	1.11	2.79e-3	1.70	1.43	1.48	2.38	1
2048	8.03e-4	1.89	1.13e-3	1.30	1.41	1.70	1.41	0
8192	3.73e-4	1.10	3.49e-4	1.70	1.43	1.48	2.38	1

Table 13: E2, $r^{3/2}$, $x_0 = \frac{1}{3}$, BDM_2, uniform refinement. r: rate with respect to h.

$\#\mathcal{T}_h$	$\|\boldsymbol{q}-\boldsymbol{q}_h\|$	r	$\|\nabla_h(u-\nu_h)\|$	r	$\|u-\nu_h\|$	r	$\frac{\eta_{\min}}{\mathbb{E}}$	$\frac{\eta_{\mathrm{FHS}}}{\mathbb{E}}$	$\frac{\mathrm{osc}}{\mathbb{E}}$	$\frac{(\eta^2+\mathrm{osc}^2)^{1/2}}{\mathbb{E}}$
8	2.53e-1	–	2.84e-1	–	3.48e-2	–	1.07	1.26	1.58	1.91
32	2.93e-2	3.11	3.08e-2	3.20	2.43e-3	3.84	0.95	1.39	1.71	1.96
128	3.65e-3	3.00	3.71e-3	3.05	1.44e-4	4.08	0.81	2.24	1.74	1.92
512	4.68e-4	2.97	4.67e-4	2.99	8.27e-6	4.12	0.72	4.40	1.71	1.86
2048	5.96e-5	2.97	5.91e-5	2.98	4.90e-7	4.08	0.67	8.74	1.69	1.82
8192	7.55e-6	2.98	7.46e-6	2.99	2.99e-8	4.03	0.65	17.41	1.67	1.79

Table 14: E3, FHS 4.2 with BDM_2 and Stenberg's full constraint. r: rate with respect to h.

$\#\mathcal{T}_h$	$r^{7/2}$, $x_0 = \frac{1}{3}$				FHS 4.2			
	δ_∇	δ_{L^2}	n_{act}	$\min(\nu_h^0 - \chi)$	δ_∇	δ_{L^2}	n_{act}	$\min(\nu_h^0 - \chi)$
8	1.46e-1	4.15e-2	1	-4.70e-2	1.21e-1	2.17e-2	3	-5.34e-2
32	6.57e-1	9.28e-1	2	-1.67e-3	7.04e-2	8.61e-3	6	-3.74e-3
128	9.09e-2	1.77e-2	3	-4.11e-4	3.45e-2	2.55e-3	12	-2.49e-4
512	2.03e-1	3.10e-1	6	-1.31e-5	1.63e-2	2.14e-4	24	-1.65e-5
2048	3.20e-2	4.87e-3	11	-3.24e-6	7.83e-3	3.49e-4	48	-1.08e-6
8192	5.48e-2	8.69e-2	22	-1.02e-7	3.81e-3	3.20e-4	96	-6.94e-8
32768	8.83e-3	1.25e-3	43	-2.54e-8				

Table 15: E5 with BDM_2: same quantities as in Table 8.

$\#\mathcal{T}_h$	$k=1$ (RT_0)				$k=2$ (BDM_1)				$k=3$ (BDM_2)			
	$\mathbb{E}$	r	$\frac{\eta}{\mathbb{E}}$	$\frac{\mathrm{osc}}{\mathbb{E}}$	$\mathbb{E}$	r	$\frac{\eta}{\mathbb{E}}$	$\frac{\mathrm{osc}}{\mathbb{E}}$	$\mathbb{E}$	r	$\frac{\eta}{\mathbb{E}}$	$\frac{\mathrm{osc}}{\mathbb{E}}$
4	2.10e+0	–	0.43	3.46	1.96e+0	–	0.24	3.23	1.91e+0	–	0.24	3.55
16	1.97e+0	0.09	0.49	1.64	1.65e+0	0.25	0.17	1.89	1.06e+0	0.86	0.29	2.41
64	1.29e+0	0.61	0.68	0.83	9.16e-1	0.85	0.32	1.18	2.11e-1	2.32	0.37	2.16
256	8.35e-1	0.63	0.77	0.34	2.58e-1	1.83	0.28	1.12	4.27e-2	2.31	0.52	1.97
1024	4.32e-1	0.95	0.77	0.18	7.49e-2	1.78	0.33	1.05	6.82e-3	2.65	0.68	1.52
4096	2.19e-1	0.98	0.76	0.09	1.94e-2	1.95	0.37	1.03	1.64e-3	2.06	0.86	0.80
16384	1.10e-1	0.99	0.75	0.05	5.04e-3	1.94	0.43	0.99				

Table 16: E6, BGK 7.2, uniform refinement. r: rate of $\mathbb{E}$ with respect to h.

ex.	k	$\#\mathcal{T}_h$	defect	η_{rec}/A	$(\eta_{\mathrm{rec}}^2 + 2\lvert(r,e)\rvert)^{1/2}/A$	$\frac{\lVert\nabla_h(\nu_h - v_h)\rVert}{J}$
FHS 4.2	2	8	0.00e+0	0.78	1.01	1.67
FHS 4.2	2	32	2.24e-16	0.69	1.01	1.79
FHS 4.2	2	128	1.02e-15	0.67	1.01	1.76
FHS 4.2	2	512	1.39e-15	0.66	1.01	1.71
FHS 4.2	2	2048	2.31e-14	0.66	1.01	1.67
FHS 4.2	3	8	6.59e-16	0.88	1.00	2.00
FHS 4.2	3	32	6.74e-15	0.81	1.00	2.17
FHS 4.2	3	128	1.08e-13	0.72	1.00	2.33
FHS 4.2	3	512	1.53e-12	0.68	1.00	2.45
FHS 4.2	3	2048	1.89e-11	0.66	1.00	2.50
BGK 7.2	2	4	3.18e-2	0.18	1.02	1.88
BGK 7.2	2	16	9.79e-4	0.19	1.00	2.01
BGK 7.2	2	64	2.73e-5	0.36	1.00	1.86
BGK 7.2	2	256	1.15e-6	0.35	1.00	1.90
BGK 7.2	2	1024	1.53e-6	0.37	1.00	1.75
BGK 7.2	2	4096	2.78e-6	0.41	1.00	1.72
BGK 7.2	2	16384	5.15e-6	0.47	1.00	1.73
BGK 7.2	3	4	4.22e-2	0.23	0.98	2.48
BGK 7.2	3	16	1.14e-4	0.31	1.00	2.84
BGK 7.2	3	64	4.70e-5	0.44	1.00	2.46
BGK 7.2	3	256	1.20e-5	0.52	1.00	2.64
BGK 7.2	3	1024	3.26e-5	0.74	1.00	2.70
BGK 7.2	3	4096	6.03e-5	0.94	1.00	2.72

uniform		adaptive	
N	η	N	η
696	5.83e-2	696	5.83e-2
2736	4.03e-2	1875	7.07e-3
10848	2.57e-2	4971	1.59e-3
43200	1.67e-2	14418	3.17e-4
		36210	7.92e-5
		100056	1.69e-5
		276993	3.60e-6
$N^{-0.31}$		$N^{-1.52}$	

Table 17: Left: verification of the identity (relative defect, η_{rec}/A and empirical lower bound for $C_{\mathcal{A}}$). Right: E4, BGK 7.3 with BDM_2; fitted rates in the last row.

Observations with $k = 3$.

(9) *Oscillation with $k = 3$ (Table 10).* With $f \notin \mathbb{P}^1$, $\mathrm{osc}/\mathbb{E} \approx 2.0$ and $\eta/\mathbb{E} \approx 0.15$; with $f \in \mathbb{P}^1$, $\eta/\mathbb{E} \to 0.79$. On the initial mesh with $f \notin \mathbb{P}^1$, $\eta \approx 10^{-7}$ while $\mathbb{E} = 0.18$: this is exactly the situation of Remark 6.3, where ν_h solves the problem with data $Q_h f$.

(10) *Superconvergence with $k = 3$ (E1, Table 11).* With the transition away from the nodes, the rates of $\boldsymbol{q}_h$ and $\nabla_h \nu_h$ oscillate around 3. Those of ν_h in L^2 and of $Q_h u - u_h$ alternate between two families, according to the relative position of x_0 in the edge, which alternates between two values under bisection; within each family the rate is 4 (for instance $4.70 \cdot 10^{-5}$, $1.85 \cdot 10^{-7}$ and $7.25 \cdot 10^{-10}$ on the even levels). Full superconvergence is preserved also for $k = 3$.

(11) *Contact term with $k = 3$ (Tables 11, 12 and 13).* $\eta_{\min}/\mathbb{E}$ stays in $[0.87, 1.19]$ with $x_0 = \frac{1}{3}$ and in $[0.94, 1.04]$ with $x_0 = 0$; the alternation with $x_0 = \frac{1}{3}$ comes from the volume part, not from the contact term. $\eta_{\mathrm{FHS}}/\mathbb{E}$ doubles at each level (up to 26 and 23), as $h^{-(k-1)/2} = h^{-1}$ predicts. In E2, at the levels where the rule chooses complementarity on the transition edge, $\eta_{\mathrm{tr}}/\mathbb{E} = 2.38$ against 1.43 for $\eta_{\min}$: with $k = 3$ the minimum rule matters more.

(12) *Singularity and adaptivity with $k = 3$ (Tables 13 and 12).* On uniform meshes, E2 alternates around the limit $h^{3/2}$ and the effectivity is constant (1.41–1.43). With adaptive refinement, $\mathbb{E} \sim N^{-1.52}$, the optimal rate $N^{-3/2}$, with stable effectivity 0.85.

(13) *FHS 4.2 with $k = 3$ (Table 14).* With Stenberg's full constraint, the rates are 3 for $\boldsymbol{q}_h$ and $\nabla_h \nu_h$ and 4 in L^2. Since $f \notin \mathbb{P}^1$, $\mathrm{osc}/\mathbb{E} \approx 1.7$ and $\eta_{\min}/\mathbb{E}$ decreases to 0.65, while $(\eta^2 + \mathrm{osc}^2)^{1/2}/\mathbb{E} \approx 1.8$ is stable.

(14) *Bernstein constraint with $k = 3$ (Table 15).* Unlike $k = 2$, the effect is of order one on coarse meshes: up to 66 % in the gradient and 93 % in L^2 with $r^{7/2}$, on one of the two kinds of level,

according to the relative position of the transition in its edge. There $u \sim r^{7/2}$ vanishes to high order, and its best degree-3 approximation may have negative Bernstein coefficients. The effect is not always adverse (on some levels the constrained error is smaller) and it decays with h: on the finest mesh it is 0.9 % with $r^{7/2}$ and 0.4 % in FHS 4.2. It does not change the rates.

(15) *BGK 7.2 (Table 16).* With the transition at a node, $k = 1$ gives rate 1 and effectivity 0.75; $k = 2$ gives a rate close to 2. Here f has large gradients, so osc is comparable to or larger than the error: $\mathrm{osc}/\mathbb{E} \approx 1$ for $k = 2$ and between 3.6 and 0.8 for $k = 3$, while $(\eta^2 + \mathrm{osc}^2)^{1/2}/\mathbb{E}$ ends at 1.08 and 1.18. For $k = 3$ the rate drops from 2.65 to 2.06 on the last level, on its way to the limit $h^{3/2}$ of the singularity.

(16) *Verification of the identity (Table 17).* In FHS 4.2, with polynomial data, the relative defect is 10^{-16} to 10^{-11}: the identity holds to machine precision. In BGK 7.2 it is 10^{-6} to 10^{-5}, the quadrature error for singular functions. η_{rec}/A ranges from 0.18 to 0.94, but $(\eta_{\mathrm{rec}}^2 + 2|(r, e)|)^{1/2}/A = 1.00$: everything η_{rec} does not see is the data term. The empirical lower bound for $C_{\mathcal{A}}$ is 1.7–2.0 for $k = 2$ and 2.0–2.8 for $k = 3$.

(17) *BGK 7.3 with $k = 3$ (Table 17).* Uniform: $\eta \sim N^{-0.31}$; adaptive: $\eta \sim N^{-1.52}$, the optimal rate. The contact term is 1.5 % of η at the end. The adaptive mesh reaches elements of size $\sim 10^{-8}$ next to the Dirichlet–Neumann junction, and there the direct solver loses accuracy unless the system is equilibrated by rows and columns: without equilibration, spurious jumps appeared in η.

(18) *Solver.* Starting the active set from the classification of the previous level, the primal-dual method terminated in at most 8 iterations for $k = 3$, with zero KKT residual.

11 Why these terms and not others

(a) *The FHS term $\langle \gamma_n u_h, \gamma_0 u_h \rangle_E$ [4], in dual-mixed form.*

- With the method's own trace, the multiplier: by Proposition 2.1, $\sum_i \mu_{E,i} b_{E,i}(\boldsymbol{q}_h \cdot \boldsymbol{n}) \equiv 0$ and the term carries no information.
- With the postprocessed trace, in the form $(v_h^0, -\boldsymbol{q}_h \cdot \boldsymbol{n})_E$ that enters (10): the term is reliable, but it is linear in the error. On contact edges, $-\boldsymbol{q}_h \cdot \boldsymbol{n} = O(1)$ and $\nu_h = O(h^{k+1})$. The *order count* gives $\sum_E \sim h^{k+1}$, whereas the squared energy error is $\sim h^{2k}$. The orders coincide for $k = 1$ (the FHS regime) and the term dominates for $k \geq 2$, which includes all of BDM_p. Tables 3, 4 and 12 confirm this: $\eta_{\mathrm{FHS}}/\mathbb{E}$ grows like $h^{-(k-1)/2}$, that is, by a factor close to $\sqrt{2}$ per level for $k = 2$ and close to 2 for $k = 3$.
- The minimum rule in (7) uses this term only where it is cheaper than zeroing the trace, typically at transitions.

(b) *Positivity.* With (H2), no positivity term is needed. Without (H2), the Bernstein-coefficient version, $\sum_i ((c_i)_-)^2$, is reliable but, for $k \geq 2$, not efficient, because a coefficient can be negative while $\nu_h \geq 0$. The L^2 version, $h_E^{-1} \|(\nu_h)_-\|_E^2$, is efficient, but I have no proof of reliability: a negative sliver of height δ and width $t \ll h$ costs the reconstruction $\sim \delta$ in energy, while the term only sees $\delta(t/h)^{1/2}$. This is why the constraint is imposed in the postprocessing. Moreover, Lemma 3.1(e) shows that its cost ends up inside $\tilde{\eta}_K$, and Table 8 shows that for $k = 2$ it does not change the error. For $k = 3$ (Table 15) the effect is of order one on some levels, because the Bernstein condition is stronger than positivity near the transition.

(c) $\boldsymbol{q}_h \cdot \boldsymbol{n}$. It needs no term of its own. Its sign is exact by (H1), outside $\mathcal{C}_h$ it vanishes by definition, and its error on $\mathcal{C}_h$ is controlled by $\|\boldsymbol{q} - \boldsymbol{q}_h\|$ in (10). If the sign is not exact, the local correction of Remark 2.4 suffices.

(d) *Discarded variant.* Imposing $c_\alpha = 0$ on I_E for $E \in \mathcal{C}_h$ sets ν_h to zero on the whole edge, including where a transition lies inside E. It has exactly the defect that the minimum rule corrects.

12 Open items

1. A local bound for $\eta_{C,E}$ on transition edges. The situation is comparable to the primal case (Remark 8.3).
2. Convergence of the multipliers of Proposition 2.1, which would justify that the misclassified edges are only the transition ones (Remark 2.2).
3. A priori superconvergence of ν_h under the Bernstein constraint. There is classical a priori analysis of mixed methods for variational inequalities [6], but I know of no supercloseness result for the dual mixed Signorini problem. The experiments with $k = 2$ and $k = 3$ show optimal rates.
4. A less conservative positivity constraint for $k \geq 3$, for instance on the Bernstein coefficients after degree elevation, which would reduce the cost observed near transitions. The obstacle is that the vertex averaging in the reconstruction does not preserve that condition.
5. Robustness in c.
6. BDM_p with $p \geq 3$ and unstructured meshes.

Code. The prototype (`sigmrv.py`, `sigmrv2.py`, `examples.py`, `run_all.py`, `run_all3.py`, `make_tables.py`, `make_tables3.py`) accompanies this note and reproduces all tables.

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